

How to calculate the current-current correlation function, correct to leading order of nonlinearity

Jian-Sheng Wang

Department of Physics and Center for Computational Science and Engineering,
National University of Singapore, Singapore 117542, Republic of Singapore

(Dated: 23 December 2010)

This note gives the key steps and formulae for the calculation of current-current correlation function, using a theory discussed by W. Götze and K. H. Michel, “Self-consistent phonons,” in *Dynamical Properties of Solids*, edited by G. K. Horton and A. A. Maradudin, North-Holland, 1974. The main advantage over other methods, such as Boltzmann equation, or single-mode relaxation kinetic theory, is that we can claim that the result is exact to leading order of nonlinear interaction.

PACS numbers:

I. INTRODUCTION

The basic steps are easily stated. First, both the Hamiltonian H and associated evolution operator $\mathcal{H} = (1/\hbar)[H, \cdot]$ and the energy density operator in momentum space $e(\mathbf{q})$ are separated as a linear and nonlinear part.

$$\mathcal{H} = \mathcal{H}^0 + \mathcal{H}', \quad (1)$$

$$e(\mathbf{q}) = e^0(\mathbf{q}) + e'(\mathbf{q}). \quad (2)$$

Then the current operator $\mathbf{J}$ is given by

$$\mathcal{H} e(\mathbf{q}) = -\mathbf{J} \cdot \mathbf{q} + O(q^2). \quad (3)$$

The bold face signifies a 3D vector, and the dot $(\cdot)$ is for scalar product. we also split $\mathbf{J}$ into a linear and nonlinear part, given

$$-\mathbf{J}^0 \cdot \mathbf{q} + O(q^2) = \mathcal{H}^0 e^0, \quad (4)$$

$$-\mathbf{J}' \cdot \mathbf{q} + O(q^2) = \mathcal{H}^0 e' + \mathcal{H}' e^0. \quad (5)$$

The current-current correlation function can be computed from the correlation of

$$\mathcal{H}\mathbf{J} = \mathcal{H}^0\mathbf{J}^0 + \mathcal{H}'\mathbf{J}^0 + \mathcal{H}^0\mathbf{J}' + O(\alpha^2), \quad (6)$$

where α is the strength of nonlinear interaction. The purpose of this note is to give explicit formulas for $\mathbf{J}$, $\mathcal{H}\mathbf{J}$, and the correlation function, for a very general lattice model and also formulas in special cases of the FPU α and β model.

II. THE GENERAL MODEL

We assume each unit cell ℓ contains a certain number of atoms labelled by i and the cartesian component x (or y , or z); the last two labels will be combined as $s = (i, x)$. The relative displacement away from equilibrium position at zero temperature multiplied by $\sqrt{m_i}$ will be called u_ℓ^s , as usual. The Hamiltonian $H = H^0 + H'$ is then

$$H^0 = \frac{1}{2} \sum_{\ell, s} p_{\ell, s}^2 + \frac{1}{2} \sum_{\ell\ell', ss'} K_{\ell\ell'}^{ss'} u_\ell^s u_{\ell'}^{s'}, \quad (7)$$

$$H' = \frac{1}{3} \sum_{\ell\ell'\ell'', ss's''} T_{\ell\ell'\ell''}^{ss's''} u_\ell^s u_{\ell'}^{s'} u_{\ell''}^{s''}. \quad (8)$$

The local energy $e_\ell = e_\ell^0 + e'_\ell$ associated with cell ℓ will be defined as

$$e_\ell^0 = \frac{1}{2} \sum_s p_{\ell, s}^2 + \frac{1}{2} \sum_{\ell', ss'} K_{\ell\ell'}^{ss'} u_\ell^s u_{\ell'}^{s'}, \quad (9)$$

$$e'_\ell = \frac{1}{3} \sum_{\ell'\ell'', ss's''} T_{\ell\ell'\ell''}^{ss's''} u_\ell^s u_{\ell'}^{s'} u_{\ell''}^{s''}. \quad (10)$$

It is clear that $H = \sum_\ell e_\ell$. We'll see later that the current operator depends on the explicit definition of local energy. By implication of Green-Kubo formula, it means that the thermal conductivity also depends on how the local energies are defined. This does not seem to me right.

The wave-vector dependent energy is defined by Fourier transform in space:

$$e(\mathbf{q}) = \frac{1}{\sqrt{N}} \sum_\ell e_\ell \exp(-i\mathbf{q} \cdot \mathbf{R}_\ell), \quad (11)$$

where N is the total number of unit cells and $\mathbf{R}_\ell$ is real space Bravais lattice vector.

We like to work in normal mode coordinates Q_k and P_k , where the index $k = (\mathbf{k}, \sigma)$ stands for the wave vector and phonon mode. In order to do this, we need to solve the standard lattice dynamics problem, i.e., the eigenvalue problem:

$$D\epsilon = \omega^2 \epsilon, \quad (12)$$

where the dynamic matrix is

$$D(\mathbf{q}) = \frac{1}{N} \sum_{\ell\ell'} K_{\ell\ell'} \exp[i\mathbf{q} \cdot (\mathbf{R}_{\ell'} - \mathbf{R}_\ell)]. \quad (13)$$

Since the dynamic matrix is hermitian, $D^{s's}(\mathbf{q})^* = D^{ss'}(\mathbf{q})$, the eigenvalues ω^2 are real; we also have $D^*(\mathbf{q}) = D(-\mathbf{q})$ by definition. Assuming this is done, we get the polarization vector ϵ labelled by the wave-vector

$\mathbf{q}$ and branch index σ . We combine this two indices as $q = (\mathbf{q}, \sigma)$. We assume

$$\epsilon_\sigma^*(\mathbf{q}) = \epsilon_\sigma(-\mathbf{q}), \quad (14)$$

$$\epsilon_\sigma^\dagger \epsilon_{\sigma'} = \delta_{\sigma, \sigma'}, \quad (15)$$

$$\sum_\sigma \epsilon_\sigma^s \epsilon_\sigma^{s'*} = \delta_{s, s'}. \quad (16)$$

That is, ϵ forms an orthogonal and complete set for each given $\mathbf{q}$. The $\delta_{\sigma, \sigma'}$ is the Kronecker delta function. We also like to introduce the notation $\delta(q - k)$ where $q = (\mathbf{q}, \sigma)$, $k = (\mathbf{k}, \tau)$, which will be defined to be 1, if $\mathbf{q} - \mathbf{k}$ equals a reciprocal lattice vector $\mathbf{K}$ and $\sigma = \tau$, and zero otherwise. $-k$ is understood to be $(-\mathbf{k}, -\sigma)$ and is equivalent to $(-\mathbf{k}, \sigma)$. Note that we consider σ and $-\sigma$

as exactly the same.

The displacement can be expanded in normal modes as

$$u_\ell = \frac{1}{\sqrt{N}} \sum_k Q_k \epsilon_k \exp(i\mathbf{k} \cdot \mathbf{R}_\ell). \quad (17)$$

Both u_ℓ and ϵ_k are vectors with components indexed with s , k means the vector $\mathbf{k}$ and branch index σ . We assume a finite system with N unit cells with periodic boundary condition. In such a case, the $\mathbf{k}$ is discretized in the first BZ. Since u is hermitian, we must require $Q_k^\dagger = Q_{-k}$. Also by convention, $\omega_k = \omega_{-k} \geq 0$. Using the transformation, Eq.(17), the Hamiltonian can be written as

$$H^0 = \frac{1}{2} \sum_{k, k'} \delta(k + k') \{ P_k P_{k'} + \omega_k^2 Q_k Q_{k'} \}, \quad (18)$$

$$H' = \frac{1}{3} \sum_{kk'k''} T(kk'k'') Q_k Q_{k'} Q_{k''}; \quad (19)$$

$$T(kk'k'') = N^{-3/2} \sum_{\ell\ell'\ell'', ss's''} T_{\ell\ell'\ell''}^{ss's''} \epsilon_k^s \epsilon_{k'}^{s'} \epsilon_{k''}^{s''} e^{i(\mathbf{k} \cdot \mathbf{R}_\ell + \mathbf{k}' \cdot \mathbf{R}_{\ell'} + \mathbf{k}'' \cdot \mathbf{R}_{\ell''})}. \quad (20)$$

Note the T is completely symmetric with respect to the permutation of indices, both for the real space version (l, s) and wavevector (k) version. Due to lattice translational symmetry, one can show that

$$T(kk'k'') \propto \delta(\mathbf{k} + \mathbf{k}' + \mathbf{k}''). \quad (21)$$

One should pay attention of my notation on delta-functions: $\delta(k)$ is a 4-dimensional delta in vector and mode, but $\delta(\mathbf{k})$ is 3D vector ($\mathbf{k} = 0$ modulo the reciprocal lattice vectors). Using the definition (11) for $e(\mathbf{q})$ we can write the local energy in momentum space as

$$e(\mathbf{q}) = \frac{1}{2} \sum_{kk'} A(\mathbf{q}, kk') P_k P_{k'} + \frac{1}{2} \sum_{kk'} B(\mathbf{q}k, k') Q_k Q_{k'} + \frac{1}{3} \sum_{kk'k''} T(\mathbf{q}k, k'k'') Q_k Q_{k'} Q_{k''}, \quad (22)$$

$$A(\mathbf{q}, kk') \equiv N^{-1/2} \sum_s \epsilon_k^s \epsilon_{k'}^s \delta(\mathbf{k} + \mathbf{k}' - \mathbf{q}), \quad (23)$$

$$B(\mathbf{q}k, k') \equiv N^{-3/2} \sum_{\ell\ell', ss'} K_{\ell\ell'}^{ss'} \epsilon_k^s \epsilon_{k'}^{s'} e^{i(\mathbf{k} - \mathbf{q}) \cdot \mathbf{R}_\ell + i\mathbf{k}' \cdot \mathbf{R}_{\ell'}}, \quad (24)$$

$$T(\mathbf{q}k, k'k'') \equiv N^{-3/2} \sum_{\ell\ell'\ell'', ss's''} T_{\ell\ell'\ell''}^{ss's''} \epsilon_k^s \epsilon_{k'}^{s'} \epsilon_{k''}^{s''} e^{i(\mathbf{k} - \mathbf{q}) \cdot \mathbf{R}_\ell + i\mathbf{k}' \cdot \mathbf{R}_{\ell'} + i\mathbf{k}'' \cdot \mathbf{R}_{\ell''}}. \quad (25)$$

$T(\mathbf{q}k, k'k'')$ is symmetric only about the permutation of the last two arguments.

III. CURRENT OPERATOR

To get an expression for the current operator, we need to do in two steps, first, the commutator $[H, e(\mathbf{q})]$ need to be worked out. Then we expand the result for small $\mathbf{q}$, the negative of linear coefficient gives $\mathbf{J}$. The basic

starting point is the canonical commutation relation, in our notation, as

$$[Q_q, P_k] = i\hbar\delta(q+k). \quad (26)$$

The following formulas will be helpful:

$$\begin{aligned} \frac{1}{\hbar}[H^0, Q_k] &= -iP_k, \\ \frac{1}{\hbar}[H^0, P_k] &= i\omega_k^2 Q_k. \end{aligned} \quad (27)$$

Carrying out the commutator computation, we find

$$\begin{aligned} \mathcal{H}^0 e^0 &= \frac{i}{2} \sum_{kk'} A(\mathbf{q}, kk') \{ \omega_k^2 Q_k P_{k'} + \omega_{k'}^2 P_k Q_{k'} \} \\ &+ \frac{(-i)}{2} \sum_{kk'} B(\mathbf{q}k, k') \{ P_k Q_{k'} + Q_k P_{k'} \}. \end{aligned} \quad (28)$$

We can regroup the terms. The coefficient of $Q_k P_{k'}$ is $(i/2)\{A(\mathbf{q}, kk')\omega_k^2 - B(\mathbf{q}k, k')\}$, and the coefficient of the $P_k Q_{k'}$ is $(i/2)\{A(\mathbf{q}, kk')\omega_{k'}^2 - B(\mathbf{q}k, k')\}$. Using the definitions for A and B , and the dynamic matrix D and its hermitian and $\mathbf{k}$ inversion properties, and the fact $D\epsilon = \omega^2\epsilon$, as well as the translational invariance properties of $K_{\ell\ell'}$, we can show the second coefficient is 0, and

$$\mathcal{H}^0 e^0 = \frac{i}{2\sqrt{N}} \sum_{kk'} \epsilon_k^T \{ D(\mathbf{k}' - \mathbf{q}) - D(\mathbf{k}') \} \epsilon_{k'} \delta(\mathbf{k} + \mathbf{k}' - \mathbf{q}) Q_k P_{k'}, \quad (29)$$

where ϵ is understood to be a column vector, and the superscript T standard for vector or matrix transpose. Now we can take the limit $\mathbf{q} \rightarrow 0$. Comparing with the definition of $\mathbf{J}^0$, we find

$$\mathbf{J}^0 = \sum_{kk'} \mathbf{E}(kk') Q_k P_{k'}, \quad (30)$$

$$\mathbf{E}(kk') \equiv \frac{i}{2\sqrt{N}} \epsilon_k^T \{ \nabla_{\mathbf{k}'} D(\mathbf{k}') \} \epsilon_{k'} \delta(\mathbf{k} + \mathbf{k}'). \quad (31)$$

Note that $\mathbf{E}$ is antisymmetric with respect to the permutation of the two arguments.

We now look for the nonlinear contribution $\mathbf{J}'$ based on $\mathcal{H}' e^0 + \mathcal{H}^0 e'$. We note that

$$\frac{1}{\hbar} [Q_k, e^0(\mathbf{q})] = i \sum_p A(\mathbf{q}, (-k)p) P_p. \quad (32)$$

Using this result, we obtain

$$\begin{aligned} \mathcal{H}' e^0(\mathbf{q}) &= \frac{1}{\hbar} \left[\frac{1}{3} \sum_{kk'k''} T(kk'k'') Q_k Q_{k'} Q_{k''}, e^0 \right] \\ &= \frac{i}{3} \sum_{kk'k'', p} T(kk'k'') \left\{ A(\mathbf{q}, (-k)p) P_p Q_{k'} Q_{k''} \right. \\ &\quad \left. + A(\mathbf{q}, (-k')p) Q_k P_p Q_{k''} \right. \\ &\quad \left. + A(\mathbf{q}, (-k'')p) Q_k Q_{k'} P_p \right\}. \end{aligned} \quad (33)$$

Similarly, we have

$$\begin{aligned} \mathcal{H}^0 e'(\mathbf{q}) &= \frac{1}{\hbar} [H^0, \frac{1}{3} \sum_{kk'k''} T(\mathbf{q}, kk'k'') Q_k Q_{k'} Q_{k''}] \\ &= -\frac{i}{3} \sum_{kk'k'', p} T(\mathbf{q}, kk'k'') \left\{ P_k Q_{k'} Q_{k''} \right. \\ &\quad \left. + Q_k P_{k'} Q_{k''} + Q_k Q_{k'} P_{k''} \right\}. \end{aligned} \quad (34)$$

Swapping the index p in the first term back to k, k' or k'' , respectively, and combining the two terms, we have

$$\begin{aligned} \mathcal{H}' e^0 + \mathcal{H}^0 e' &= \frac{i}{3} \sum_{kk'k''} \left\{ \left[\left(\sum_p T(pk'k'') A(\mathbf{q}, (-p)k) \right) - T(\mathbf{q}, kk'k'') \right] P_k Q_{k'} Q_{k''} \right. \\ &\quad \left. + \left[\left(\sum_p T(kpk'') A(\mathbf{q}, (-p)k') \right) - T(\mathbf{q}, kk'k'') \right] Q_k P_{k'} Q_{k''} \right. \\ &\quad \left. + \left[\left(\sum_p T(kk'p) A(\mathbf{q}, (-p)k'') \right) - T(\mathbf{q}, kk'k'') \right] Q_k Q_{k'} P_{k''} \right\}. \end{aligned} \quad (35)$$

Now we use the explicit expressions for T and A . The TA term contains five polarization vectors, ϵ . Using the complete relation after summing over σ , we can reduce them to three. Using also the translational invariance

$T_{\ell\ell'\ell''} = T_{0,(\ell'-\ell),(\ell''-\ell)}$ or similar ones, we can factor out a common term $\delta(\mathbf{k} + \mathbf{k}' + \mathbf{k}'' - \mathbf{q})$, and finally taking the $\mathbf{q} \rightarrow 0$ limit, we obtain

$$\mathbf{J}' = \sum_{kk'k''} \left\{ \mathbf{F}^{100}(kk'k'')P_k Q_{k'} Q_{k''} + \mathbf{F}^{010}(kk'k'')Q_k P_{k'} Q_{k''} + \mathbf{F}^{001}(kk'k'')Q_k Q_{k'} P_{k''} \right\} \quad (36)$$

$$= \sum_{kk'k''} \left\{ \mathbf{F}^{010}(kk'k'') + \mathbf{F}^{010}(k''k'k) \right\} Q_k P_{k'} Q_{k''}; \quad (37)$$

$$\mathbf{F}^{100}(kk'k'') = 0, \quad (38)$$

$$\mathbf{F}^{010}(kk'k'') = -\frac{1}{3} \frac{1}{N^{3/2}} \sum_{ss's'', kk'k''} T_{kk'k''}^{ss's''} \epsilon_k^s \epsilon_{k'}^{s'} \epsilon_{k''}^{s''} (\mathbf{R}_{\ell'} - \mathbf{R}_{\ell}) e^{i(\mathbf{k} \cdot \mathbf{R}_{\ell} + \mathbf{k}' \cdot \mathbf{R}_{\ell'} + \mathbf{k}'' \cdot \mathbf{R}_{\ell''})}, \quad (39)$$

$$\mathbf{F}^{001}(kk'k'') = -\frac{1}{3} \frac{1}{N^{3/2}} \sum_{ss's'', kk'k''} T_{kk'k''}^{ss's''} \epsilon_k^s \epsilon_{k'}^{s'} \epsilon_{k''}^{s''} (\mathbf{R}_{\ell''} - \mathbf{R}_{\ell}) e^{i(\mathbf{k} \cdot \mathbf{R}_{\ell} + \mathbf{k}' \cdot \mathbf{R}_{\ell'} + \mathbf{k}'' \cdot \mathbf{R}_{\ell''})}. \quad (40)$$

We note $\mathbf{F}^{010}$ is antisymmetric with respect to the permutation of the first two arguments, $\mathbf{F}^{010}(kk'k'') = -\mathbf{F}^{010}(k'kk'')$, and $\mathbf{F}^{001}$ is antisymmetric with respect to the permutation of the first and last index. The two $\mathbf{F}$ s are related by $\mathbf{F}^{001}(kk'k'') = \mathbf{F}^{010}(kk''k')$, and also $\mathbf{F}^{001}(kk'k'')^* = \mathbf{F}^{001}(-k, -k', -k'')$. With these known relations, in Eq. (36), we can show that the second term and last term are hermitian conjugate of each other, and $\mathbf{J}'$ is a hermitian operator. We can use either the last or middle term plus its conjugate, but I prefer to use the QPQ term as it can be combined into a single term as given by Eq. (37). The expressions for the currents, Eq. (30) and (36) are consistent with that obtained by Hardy in Phys. Rev. **132**, 168 (1963).

IV. COMPUTE $[\mathbf{H}, \mathbf{J}]$

We can acting $\mathcal{H}$ on $\mathbf{J}$:

$$\mathcal{H}\mathbf{J} = \mathcal{H}^0\mathbf{J}^0 + \mathcal{H}'\mathbf{J}^0 + \mathcal{H}^0\mathbf{J}' + O(\alpha^2). \quad (41)$$

Due to the off-diagonal terms in modes, $\mathcal{H}^0\mathbf{J}^0$ is nonzero. We have

$$\begin{aligned} \mathcal{H}^0\mathbf{J}^0 &= i \sum_{kk'} \mathbf{E}(kk') (-P_k P_{k'} + \omega_k^2 Q_k Q_{k'}) \\ &= -\frac{i}{2} \sum_{kk'} (\omega_k^2 - \omega_{k'}^2) \mathbf{E}(kk') Q_k Q_{k'}. \end{aligned} \quad (42)$$

In the last step, we have symmetrized the expression by adding a permuted term and divided by 2. Due the anti-symmetric property of $\mathbf{E}$ the PP term drops. The second term contribution is

$$\begin{aligned} \mathcal{H}'\mathbf{J}^0 &= \frac{1}{\hbar} \left[\frac{1}{3} \sum_{kk'k''} T(kk'k'') Q_k Q_{k'} Q_{k''}, \mathbf{J}^0 \right] \\ &= \frac{i}{3} \sum_{kk'k'', p} T(kk'k'') \left\{ \mathbf{E}(p(-k)) Q_p Q_{k'} Q_{k''} \right. \\ &\quad \left. + \mathbf{E}(p(-k')) Q_k Q_p Q_{k''} \right. \\ &\quad \left. + \mathbf{E}(p(-k'')) Q_k Q_{k'} Q_p \right\} \end{aligned}$$

$$= \sum_{kk'k''} \mathbf{S}(kk'k'') Q_k Q_{k'} Q_{k''}; \quad (43)$$

$$\begin{aligned} \mathbf{S}(kk'k'') &\equiv \frac{i}{3} \sum_p T(pk'k'') \mathbf{E}(k, -p) \\ &\quad + T(kpk'') \mathbf{E}(k', -p) \\ &\quad + T(kk'p) \mathbf{E}(k'', -p). \end{aligned} \quad (44)$$

The last equation defines $\mathbf{S}$. It is completely symmetric with respect to the permutation of indices. Using the completeness of the polarization vector, we can give a more explicit expression for the above, e.g.,

$$\begin{aligned} \sum_p T(pk'k'') \mathbf{E}(k, -p) &= \\ \frac{1}{N^{3/2}} \sum_{ss's'', \ell\ell'\ell''} T_{\ell\ell'\ell''}^{ss's''} \left(-\frac{i}{2} \nabla_{\mathbf{k}} D^{ss''}(\mathbf{k}) \epsilon_k^{s''} \right) \epsilon_{k'}^{s'} \epsilon_{k''}^{s''} \\ &\quad \times e^{i(\mathbf{k} \cdot \mathbf{R}_{\ell} + \mathbf{k}' \cdot \mathbf{R}_{\ell'} + \mathbf{k}'' \cdot \mathbf{R}_{\ell''})}. \end{aligned} \quad (45)$$

The noninteracting Hamiltonian acting on the nonlinear current operator gives

$$\begin{aligned} \mathcal{H}^0\mathbf{J}' &= \sum_{kk'k''} \tilde{\mathbf{F}}(kk'k'') \left(-i P_k P_{k'} Q_{k''} \right. \\ &\quad \left. + i \omega_{k'}^2 Q_k Q_{k'} Q_{k''} \right. \\ &\quad \left. - i Q_k P_{k'} P_{k''} \right); \end{aligned} \quad (46)$$

$$\tilde{\mathbf{F}}(kk'k'') = \mathbf{F}(kk'k'') + \mathbf{F}(k''k'k). \quad (47)$$

The QQQ term can be symmetrized with respect to the permutation of the indices, after cyclic permutation, we can replace the coefficient of $Q_k Q_{k'} Q_{k''}$ by

$$\begin{aligned} \mathbf{S}_2(kk'k'') &= \\ \frac{i}{3} \left(\tilde{\mathbf{F}}(kk'k'') \omega_{k'}^2 + \tilde{\mathbf{F}}(k'k''k) \omega_{k''}^2 + \tilde{\mathbf{F}}(k''kk') \omega_k^2 \right). \end{aligned} \quad (48)$$

Putting all the terms together, we can write

$$\begin{aligned} \mathcal{H}\mathbf{J} &= \sum_{kk'} \mathbf{G}^{00}(kk') Q_k Q_{k'} \\ &\quad + \sum_{kk'k''} \mathbf{G}^{000}(kk'k'') Q_k Q_{k'} Q_{k''} \end{aligned}$$

$$+ \sum_{kk'k''} \mathbf{G}^{010}(P_k P_{k'} Q_{k''} + Q_k P_{k'} P_{k''}) + O(\alpha^2); \quad (49)$$

$$\mathbf{G}^{00}(kk') = -\frac{i}{2} \sum_{kk'} (\omega_k^2 - \omega_{k'}^2) \mathbf{E}(kk'), \quad (50)$$

$$\mathbf{G}^{000}(kk'k'') = \mathbf{S}(kk'k'') + \mathbf{S}_2(kk'k''), \quad (51)$$

$$\mathbf{G}^{010}(kk'k'') = -i\tilde{\mathbf{F}}(kk'k''). \quad (52)$$

Note that $\mathbf{G}^{00}$ is symmetric with respect to the permutation of the two arguments, and similarly $\mathbf{G}^{000}$ is symmetric with respect to the 3! permutations of the arguments, and $\mathbf{G}^{010}$ is symmetric only with respect to permutation of the first and last argument.

If we only keep the current $\mathbf{J}^0$ due to the linear part of the Hamiltonian, then the contribution for $\mathcal{H}\mathbf{J}$ is only the $\mathbf{G}^{00}$ and $\mathbf{S}$ terms. If we use a standard definition of local energy for the FPU α or β model of the form $(u_\ell - u_{\ell+1})^n$, we find that the $\mathbf{G}^{00} = 0$ because there is one acoustic branch, and more importantly, the $\mathbf{S}$ cancels exactly the $\mathbf{S}_2$ term, given also $\mathbf{G}^{000} = 0$. This means for these models, the linear part of current operator does not contribute to the correlation function at all, in the sense above. One must consider the nonlinear contribution to the current in order to be consistent.

V. THE (RETARDED) CORRELATION FUNCTION

We need to express $\mathcal{H}\mathbf{J}$ in terms of the creation and annihilation operators as the equations of motion are simpler in terms of these. The relation between Q and P and a , $a^\dagger$ is standard:

$$Q_k = \sqrt{\frac{\hbar}{2\omega_k}} (a_k + a_{-k}^\dagger), \quad (53)$$

$$P_k = \sqrt{\frac{\hbar}{2\omega_k}} (-a_k + a_{-k}^\dagger) i\omega_k, \quad (54)$$

$$[a_k, a_p^\dagger] = \delta(k - p). \quad (55)$$

Substituting the Q and P into Eq. (49), we get

$$\begin{aligned} \mathcal{H}\mathbf{J} = & \mathbf{G}^{00} \text{ terms} \\ & + \sqrt{\frac{\hbar^3}{8\omega_k\omega_{k'}\omega_{k''}}} \left\{ \right. \\ & [\mathbf{G}^{000} + (-\omega_k - \omega_{k'})\omega_{k'}\mathbf{G}^{010}] a_k a_{k'} a_{k''} \\ & + [\mathbf{G}^{000} + (-\omega_k + \omega_{k'})\omega_{k'}\mathbf{G}^{010}] a_k a_{k'} a_{-k''}^\dagger \\ & + [\mathbf{G}^{000} + (+\omega_k + \omega_{k'})\omega_{k'}\mathbf{G}^{010}] a_k a_{-k'}^\dagger a_{k''} \\ & + [\mathbf{G}^{000} + (+\omega_k - \omega_{k'})\omega_{k'}\mathbf{G}^{010}] a_k a_{-k'}^\dagger a_{-k''}^\dagger \\ & + [\mathbf{G}^{000} + (+\omega_k - \omega_{k''})\omega_{k'}\mathbf{G}^{010}] a_{-k}^\dagger a_{k'} a_{k''} \\ & + [\mathbf{G}^{000} + (+\omega_k + \omega_{k''})\omega_{k'}\mathbf{G}^{010}] a_{-k}^\dagger a_{k'} a_{-k''}^\dagger \\ & + [\mathbf{G}^{000} + (-\omega_k + \omega_{k''})\omega_{k'}\mathbf{G}^{010}] a_{-k}^\dagger a_{-k'}^\dagger a_{k''} \\ & \left. + [\mathbf{G}^{000} + (-\omega_k - \omega_{k''})\omega_{k'}\mathbf{G}^{010}] a_{-k}^\dagger a_{-k'}^\dagger a_{-k''}^\dagger \right\} \quad (56) \end{aligned}$$

$$= \sum_{j=0}^7 \mathbf{H}_j(kk'k'') O_j(kk'k'') \quad (57)$$

We'll define the coefficients of triple creation operators as $\mathbf{H}_j$ for $j = 0$ to 7 in the order given above. Note that $\mathbf{H}_0 = \mathbf{H}_7 = \hbar^{3/2}/(8\omega_k\omega_{k'}\omega_{k''})^{1/2} [\mathbf{G}^{000} + (-\omega_k - \omega_{k''})\omega_{k'}\mathbf{G}^{010}]$, $\mathbf{H}_1 = \mathbf{H}_6$, etc. For notational simplicity, we have suppressed the arguments for $\mathbf{G}^{000}$ and $\mathbf{G}^{010}$ which always take $kk'k''$. We associate a number from 0 to 7 to each of the product of as. 0 is $O_0 = aa'a''$, 1 is $O_1 = aa'a''^\dagger$, ..., 7 is $O_7 = a^\dagger a'^\dagger a''^\dagger$.

To calculate the correlation function of the operator O_j , we use the equation of motion for any operator A and B , Eq. (1.32) in Götze and Michel:

$$z\langle\langle A; B \rangle\rangle_z = \langle[A, B]\rangle + \langle\langle \frac{1}{\hbar}[A, H]; B \rangle\rangle_z. \quad (58)$$

At this point, it is correct to replace H by H^0 . The errors introduced are high order in the nonlinear coupling α . First, some general observation on these type of correlation functions. (1) the result is 0 unless there are equal number of creation operators and annihilation operators in AB . The nonzero values are $\langle\langle 0; 7 \rangle\rangle$ (0 means O_0 , etc), $\langle\langle 1; 3 \rangle\rangle$, $\langle\langle 1; 5 \rangle\rangle$, $\langle\langle 1; 6 \rangle\rangle$, $\langle\langle 2; 3 \rangle\rangle$, $\langle\langle 2; 5 \rangle\rangle$, $\langle\langle 2; 6 \rangle\rangle$, $\langle\langle 3; 4 \rangle\rangle$, $\langle\langle 4; 5 \rangle\rangle$, $\langle\langle 4; 6 \rangle\rangle$, and another 10 with arguments swapped. (2) The creation operators always come in as a negative k . (3) If we know $\langle\langle A; B \rangle\rangle_z$, we don't need to calculate $\langle\langle B; A \rangle\rangle_z$, because they are related by $\langle\langle A; B \rangle\rangle_z = \langle\langle B; A \rangle\rangle_{-z}$. Using the equation of motion for a ,

$$\frac{1}{\hbar}[a, H^0] = \omega a, \quad (59)$$

$$\frac{1}{\hbar}[a^\dagger, H^0] = -\omega a^\dagger, \quad (60)$$

we can obtain, for example,

$$\begin{aligned} \langle\langle 0; 7 \rangle\rangle_z &= \langle\langle a_k a_{k'} a_{k''}; a_{-p}^\dagger a_{-p'}^\dagger a_{-p''}^\dagger \rangle\rangle_z \\ &= \frac{(n_k + 1)(n_{k'} + 1)(n_{k''} + 1) - n_k n_{k'} n_{k''}}{z - \omega_k - \omega_{k'} - \omega_{k''}} \\ &\quad \times [\delta(k + p)\delta(k' + p')\delta(k'' + p'') + \text{permutations}]. \quad (61) \end{aligned}$$

The product of δ functions should be permuted 3!=6 times for the second arguments of p . Where $n_k = 1/[\exp(\hbar\omega_k/k_B T) - 1]$ is the Bose distribution function. There is no point to try to commute the commutator; it is simpler just to compute directly $\langle AB \rangle - \langle BA \rangle$. The calculation is similar to applying Wick's theorem by pairing the creation operator with an annihilation operator, except that the relative order should be preserved, such that $aa^\dagger$ gets a $(n+1)$ and $a^\dagger a$ gets n . Similarly, we have

$$\begin{aligned} \langle\langle 1; 3 \rangle\rangle_z &= \langle\langle a_k a_{k'} a_{-k''}^\dagger; a_p^\dagger a_{-p'}^\dagger a_{-p''}^\dagger \rangle\rangle_z \\ &= \frac{(n_k + 1)(n_{k'} + 1)n_{k''} - n_k n_{k'}(n_{k''} + 1)}{z - \omega_k - \omega_{k'} + \omega_{k''}} \delta(k'' + p) \\ &\quad \times [\delta(k + p')\delta(k' + p'') + \delta(k + p'')\delta(k' + p')]. \quad (62) \end{aligned}$$

We have used the fact that $\omega_{-k} = \omega_k$ and $n_{-k} = n_k$. In this case, the permutations have only two choices. All the other terms have similar structure as above.

In principle, we should also consider the contraction of creation/annihilation operators within either A or B . If so the above equation is not complete. The situation becomes much simpler if we can assume that

$$Q_0 = P_0 = a_0 = a_0^\dagger = 0, \quad (63)$$

where 0 means $\mathbf{k} = 0$. Due to the translational invariance constraint, $\delta(\mathbf{k} + \mathbf{k}' + \mathbf{k}'')$ in $\mathbf{G}$ or $\mathbf{H}$ functions, if two of the internal operator pairs, the third one must take $\mathbf{k} = 0$. Using the assumption, Eq. (63), such terms will be zero. This assumption has additional simplification advantage, that is, we can commute all the a operators into a standard order, say, operators 0, 1, 3, and 7. And we only need to consider 4 correlation functions instead of 20. But, I cannot prove Eq. (63). In particular, the optical modes do not have such property.

When the correlation of the operator O_j is worked out,

the calculation of the correlation function

$$\chi_{ab}(z) = -\langle\langle \mathcal{H}J_a; \mathcal{H}J_b \rangle\rangle, \quad (64)$$

is a matter of book keeping. We have now used the component a or b of the vector $\mathbf{J}$, which takes x or y , or z . Since $\mathcal{H}\mathbf{J}$ has 8 terms (ignore the $\mathbf{G}^{00}$ for the moment, the correlation χ have $8 \times 8 = 64$ terms, but only 20 of these are nonzero. For each of the correlations $\langle\langle O_i; O_j \rangle\rangle_z$ we may have 6 or 2 terms (or more if internal pairing need to be accounted for) δ functions to be summed. If we work hard enough, we should be able to get all the terms, but we really need a good strategy to do it right (e.g., using symmetries of the problem). I'm stuck here!

Let assume that the assumption can be justified one way or other. The expressions can be greatly simplified, as we can group O_1, O_2 and O_4 together by commuting to a standard order, say O_1 , and similarly, group O_3, O_5 , and O_6 can be grouped together, and now we only need to deal with four different types of operators. The result can be expressed as

$$\begin{aligned} \chi_{ab} &= -\langle\langle \mathcal{H}J^a; \mathcal{H}J^b \rangle\rangle \\ &= -\sum_{kk'k''} \left\{ 6\check{H}_0^a(kk'k'')\check{H}_7^b(kk'k'')^* \frac{(n_k+1)(n_{k'}+1)(n_{k''}+1) - n_k n_{k'} n_{k''}}{z - \omega_k - \omega_{k'} - \omega_{k''}} \right. \\ &\quad + \check{H}_1^a(kk'k'')[\check{H}_3^b(k''kk')^* + \check{H}_3^b(k''k'k)^*] \frac{(n_k+1)(n_{k'}+1)n_{k''} - n_k n_{k'}(n_{k''}+1)}{z - \omega_k - \omega_{k'} + \omega_{k''}} \\ &\quad + \check{H}_3^a(kk'k'')[\check{H}_1^b(k''kk')^* + \check{H}_1^b(k''k'k)^*] \frac{(n_k+1)(n_{k'}+1)n_{k''} - n_k n_{k'}(n_{k''}+1)}{-z - \omega_k - \omega_{k'} + \omega_{k''}} \\ &\quad \left. \check{H}_7^a(kk'k'')\check{H}_0^b(kk'k'')^* \frac{(n_k+1)(n_{k'}+1)(n_{k''}+1) - n_k n_{k'} n_{k''}}{-z - \omega_k - \omega_{k'} - \omega_{k''}} \right\}; \end{aligned} \quad (65)$$

$$\check{\mathbf{H}}_1(kk'k'') \equiv \mathbf{H}_1(kk'k'') + \mathbf{H}_2(kk''k') + \mathbf{H}_4(k''kk'), \quad (66)$$

$$\check{\mathbf{H}}_3(kk'k'') \equiv \mathbf{H}_3(kk'k'') + \mathbf{H}_5(k'kk'') + \mathbf{H}_6(k''k'k), \quad (67)$$

$$(68)$$

where $\check{\mathbf{H}}_0$ is obtained from $\mathbf{H}_0$ by completely symmetrized the indices with respect to the $3!$ permutations, similarly for $\check{\mathbf{H}}_7$. The complex conjugate superscripts $*$ come from the fact $H_j((-k)(-k')(-k'')) = H_j^*(kk'k'')$.

If the $\mathbf{G}^{00}$ is nonzero, it creates some trouble for us. If it is zero, the error for χ is $O(\alpha^3)$ because $\mathbf{G}^{000}$ or $\mathbf{G}^{010}$ is first order in α . If the linear contribution $\mathbf{G}^{00}$ is not zero, it is 0-th order in α . Combination of $\mathbf{G}^{00}$ with higher order of α^2 which we have omitted in the current expression $\mathbf{J}$ also give second order term. I assume such calculation to keep one more order in α would be too difficult to do.

VI. χ TENSOR AND GREEN-KUBO FORMULA

In this section, we record the relevant formulas going from the tensor χ to the Green-Kubo integrand. For more detail, the reader is referred to the article of Götze and Michel and the literature on Green-Kubo formula (e.g. R. Kubo, M. Toda, and N. Hashitsume, "Statistical Physics II.").

First the inner product in the space of operators is defined as

$$\begin{aligned} (A, B) &= -\langle\langle A^\dagger; B \rangle\rangle_{z=0} = \beta \langle A^\dagger; B \rangle \\ &= \frac{i}{\hbar} \int_0^{+\infty} \langle [A(t)^\dagger, B] \rangle dt, \end{aligned} \quad (69)$$

the semicolon ‘;’ separated single angular brackets denote

Kubo correlation,

$$\langle A; B \rangle = \frac{1}{\beta} \int_0^\beta \text{Tr}(\rho e^{\lambda H} A e^{-\lambda H} B) d\lambda, \quad (70)$$

where $\beta = 1/(k_B T)$, and $\rho = \exp(-\beta H)/Z$ is the canonical distribution function. The Kubo formula takes one of the following forms:

$$\begin{aligned} \kappa_{ab} &= \frac{N}{TV\hbar} \int_0^{\hbar\beta} \int_0^{+\infty} \langle J_a(-i\lambda) J_b(t) \rangle \\ &= \frac{N}{k_B T^2 V} \int_0^{+\infty} \langle J_a; J_b(t) \rangle dt \\ &= \frac{N}{TV} \int_0^{+\infty} (J_a(t), J_b) dt. \end{aligned} \quad (71)$$

The indices a, b take cartesian components.

Let define the correlation function Φ by the matrix elements of the resolvent

$$\Phi_{ab}(z) = (J_a, (z - H)^{-1} J_b), \quad (72)$$

where z takes value on the complex plane. One can show that it is the Fourier transform (or complex Laplace transform) of the function

$$\phi_{ab}(t) = -i\theta(t)(J_a(t), J_b), \quad (73)$$

where $\theta(t)$ is the Heaviside step function. Thus the inverse Fourier transform of $\Phi(z)$ gives us the Green-Kubo integrand. The thermal conductivity is obtained by

$$k_{ab} = \frac{V}{T} i \Phi(z \rightarrow 0^+). \quad (74)$$

Finally, Φ and χ are related through:

$$\Phi(z) = [zI - m(z)x_0^{-1}]^{-1} x_0, \quad (75)$$

$$m(z) = \frac{\chi(0) - \chi(z)}{z}, \quad (76)$$

$$(x_0)_{ab} = (J_a, J_b), \quad (77)$$

where I is D -dimensional identity matrix. We still need a more accurate expression for x_0 accurate to second order in the nonlinear strength α than what is given in Götze and Michel in Eq. (38), (40). But this can be done. Since each term in χ is of the form $A_j/(z - \Omega_j)$, $m(z)$ is then a sum of the terms of the form $-A_j/[\Omega_j(z - \Omega_j)]$.

VII. FPU α MODEL

In this section we work out the detail of 1D FPU α model. Although the cubic interaction cannot support umklapp process, the mathematical structure is similar to the general model. Thus, the purpose of this section is several folds: (1) to check the general formulas derived, (2) to check if some of the assumptions are true or

not, (3) to show the definition dependence of the nonlinear current, (4) to show a miraculous cancellation that $G^{000} = 0$.

Consider the Hamiltonian,

$$\begin{aligned} H &= \frac{1}{2} \sum_{\ell=1}^N p_\ell^2 + \frac{1}{2} K \sum_{\ell=1}^N (u_\ell - u_{\ell+1})^2 \\ &\quad + \frac{1}{3} \alpha \sum_{\ell=1}^N (u_\ell - u_{\ell+1})^3. \end{aligned} \quad (78)$$

Periodic B.C. will be used, i.e., $u_\ell \equiv u_{\ell+N}$. The polarization vector $\epsilon \equiv 1$ and the normal mode representation is

$$u_\ell = \frac{1}{\sqrt{N}} \sum_k Q_k e^{ik\ell a}, \quad (79)$$

$$k = \frac{2\pi n}{Na}, \quad n = 0, 1, 2, \dots, N-1. \quad (80)$$

Since we are in 1D and there is only one phonon branch, the symbol k now will be understood to be just a number. a is the lattice constant, we'll assume $a = 1$. The dynamic matrix and the dispersion relation is given by

$$D(k) = \omega(k)^2 = 2K(1 - \cos(k)). \quad (81)$$

The nonlinear coupling T can be rewritten in one of several equivalent ways:

$$\begin{aligned} T(kk'k'') &= \frac{\alpha}{\sqrt{N}} (1 - e^{ik})(1 - e^{ik'})(1 - e^{ik''}) \delta(k + k' + k'') \\ &= \frac{8i\alpha}{\sqrt{N}} e^{i\frac{k+k'+k''}{2}} \sin\left(\frac{k}{2}\right) \sin\left(\frac{k'}{2}\right) \sin\left(\frac{k''}{2}\right) \delta(k + k' + k'') \\ &= \frac{-2i\alpha}{\sqrt{N}} [\sin(k) + \sin(k') + \sin(k'')] \delta(k + k' + k''). \end{aligned} \quad (82)$$

The δ function now stands for 1 if $k + k' + k'' = 2\pi n$, $n = 0, \pm 1, \pm 2, \dots$; and 0 otherwise.

Using either the formula, Eq. (30,31), or a direct calculation of $\mathcal{H}^0 \epsilon^0$, the linear current is

$$J^0 = -\frac{iK}{\sqrt{N}} \sum_k \sin(k) Q_k P_{-k}. \quad (83)$$

The local energy can be defined in two ways, either we use $\frac{1}{4} [(u_{\ell-1} - u_\ell)^2 + (u_\ell - u_{\ell+1})^2]$, or $\frac{1}{2} \sum_{\ell'} K_{\ell\ell'} u_\ell u_{\ell'} = \frac{1}{2} K (-u_\ell u_{\ell-1} + 2u_\ell^2 - u_\ell u_{\ell+1})$. Both definitions or a linear interpolation between the two give exactly the same J^0 . But this not the case for the nonlinear current J' . It depends on the definition. To show clearly the feature of local energy definition dependence, let consider

$$\begin{aligned} e_\ell^\lambda &= \frac{\alpha}{6} \left[(1 - \lambda)(u_{\ell-1}^3 - u_{\ell+1}^3) \right. \\ &\quad \left. + (3 + \lambda)u_{\ell-1}u_\ell^2 + (3 - \lambda)u_\ell u_{\ell+1}^2 \right. \\ &\quad \left. + (-3 + \lambda)u_{\ell-1}^2 u_\ell - (3 + \lambda)u_\ell^2 u_{\ell+1} \right]. \end{aligned} \quad (84)$$

First of all, this is a valid definition of local energy, because $\sum_{\ell} e_{\ell}^{\lambda} = H'$ independent of λ . If $\lambda = 0$ it is the same as $\frac{\alpha}{6}[(u_{\ell-1} - u_{\ell})^3 + (u_{\ell} - u_{\ell+1})^3]$. If $\lambda = 1$, it is $\frac{1}{3} \sum_{\ell'} T_{\ell\ell'\ell''} u_{\ell} u_{\ell'} u_{\ell''}$, where

$$T_{000} = 0, \quad (85)$$

$$T_{011} = T_{101} = T_{110} = 1, \quad (86)$$

$$T_{001} = T_{010} = T_{100} = -1, \quad (87)$$

$$T_{0-1-1} = T_{-10-1} = T_{-1-10} = -1, \quad (88)$$

$$T_{00-1} = T_{0-10} = T_{-100} = 1; \quad (89)$$

$$T_{\ell+m, \ell'+m, \ell''+m} = T_{\ell\ell'\ell''}. \quad (90)$$

Using this definition of local energy, we can work out the q -dependence energy as

$$e^{\lambda}(q) = \frac{i\alpha}{9\sqrt{N}} \sum_{kk'k''} \left\{ -3(1-\lambda) \sin(k+k'+k'') \right. \\ \left. -(3+\lambda) [\sin(k) + \sin(k') + \sin(k'')] \right\}$$

$$-(3-\lambda) [\sin(k'+k'') + \sin(k''+k) + \sin(k+k')] \\ \left. \right\} \times \delta(k+k'+k''-q) Q_k Q_{k'} Q_{k''}. \quad (91)$$

We can check that $\sqrt{N} e^{\lambda}(0) = H'$ is independent of λ , as expected. The nonlinear current operator can be worked out as

$$J' = \frac{i\alpha}{3\sqrt{N}} \sum_{kk'k''} \left\{ 3(1-\lambda) + (3+\lambda) \cos(k) \right. \\ \left. -(3-\lambda) [\cos(k') + \cos(k'')] \right\} \\ \delta(k+k'+k'') P_k Q_{k'} Q_{k''} + \dots \quad (92)$$

Acknowledgments

I thank Wang Jian for keeping me focused on this problem.